

## **Outline**

### **Objectives**

The objectives of this module are to enable participants:

- To obtain a thorough understanding of the logic and process of conducting surveys
- To critically evaluate surveys as a research tool
- To design and evaluate various types of sampling strategies
- To plan and undertake their own surveys

### **Themes**

- The scope of survey research and its limitations
- Surveys in health research
- Survey error
- Sampling
- Measurement and questionnaire construction
- Fieldwork and management of surveys
- Processing of data for analysis
- Ethical aspects in survey research

| # | Topic                                                                                                                                                                          |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>The scope of survey research</b><br>The nature of surveys<br>Theoretical background<br>Quantitative vs qualitative approach<br>Strengths and limitations of survey research |
| 2 | <b>Surveys in health research</b><br>Types of surveys<br>Alternative designs<br>Examples of large scale surveys                                                                |
| 3 | <b>Formalising the survey process</b><br>Define research question(s) & population<br>Steps in the survey process<br>The survey protocol                                        |
| 4 | <b>Survey error</b><br>The Total Survey Error framework<br>Understanding sources of error<br>Cost, precision and bias: a trade-off                                             |
| 5 | <b>Target population, study population and sampling</b><br>The concept of representativeness                                                                                   |

| #  | Topic                                                                                                                                                                                                          |
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| 6  | <b>Sampling strategies</b><br>Probability vs non-probability samples<br>Systematic sampling and systematic error<br>Sampling strategies and data analysis                                                      |
| 7  | <b>Power, precision and sample size</b><br>Myths about sample size calculations<br>Standard formulae are not always the right ones<br>The 'paradox' of the preliminary data<br>Cluster sampling and adjustment |
| 8  | <b>Measurement</b><br>What does it mean "to measure"?<br>Scale of measurement<br>Theoretical constructs<br>Operationalisation, indicators and scales<br>Reliability and validity                               |
| 9  | <b>Questionnaires</b><br>Types of questionnaires<br>Question content, format and wording<br>Questionnaire layout and response format                                                                           |
| 10 | <b>Non-response bias</b><br>Design vs realised sample<br>Limiting non-response bias<br>Adjusting for non-response bias                                                                                         |
| 11 | <b>Modes of administration</b><br>Self-administered/Interviewer administered<br>Digital data collection systems<br>Online surveys                                                                              |
| 12 | <b>Beyond questionnaires</b><br>Integrating self-report with biomarkers/direct measurements/observation                                                                                                        |
| 13 | <b>Data management</b><br>Principles<br>Checking, correcting, documenting<br>Data management plans<br>Data keeping and sharing                                                                                 |
| 14 | <b>Fieldwork</b><br>Planning<br>Piloting (Feasibility, Timing, Checklist)<br>Gaining access<br>Selection and training of fieldworkers<br>Monitor survey data collection and quality control                    |
| 15 | <b>Data analysis &amp; reporting</b><br>Data analysis plan and survey design<br>Analysing data from complex sampling design is... complex<br>Principles of data reporting<br>What to report: a checklist       |
| 16 | <b>Ethical Aspects</b><br>Principles<br>Risk: are all surveys low risk?<br>Informed consent<br>Participants reimbursement                                                                                      |

| #  | Topic                                                                                                                                                                      |
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| 17 | <b>Privacy and the POPI act</b><br>Survey Research and the POPI act                                                                                                        |
| 18 | <b>Qualitative methods in survey research</b><br>Qualitative research designs and data analysis methodologies for questionnaire creation<br>Analysing open-ended questions |